Mr. Davide Capone

Personal Information

Date of birth: 23/04/1999 **Nationality**: Italian

Contact Information

G.B. Morgagni 2D
Tolmezzo (UD), Italy

Phone number: (+39) 366 534 3079
E-mail: davidecapone2304@gmail.com
LinkedIn: linkedin.com/in/capone-davide/

Profile Description

I am a Data Science and Scientific Computing student with a growing foundation in artificial intelligence, data analysis, and scientific problem-solving. My academic journey began with a bachelor's degree in Internet of Things, Big Data & and Web, where I gained valuable insights into modern technologies and their applications. I am currently pursuing a master's degree in Data Science and Scientific Computing, focusing on advanced data analysis, artificial intelligence, and computational methods. I have also had the opportunity to develop my communication and presentation skills as a Computer Science teacher at a secondary school, enabling me to explain technical concepts effectively to diverse audiences. I am motivated by a genuine passion for computer science and mathematics, and I am eager to contribute to innovative AI projects, continue learning, and grow as a professional in the field.

Experience

Deep Learning Engineer (Intern)
Danieli Research Center, Buttrio (Udine), ITA 2024/12 → Present

- Researching, developing, and optimizing deep learning models for industrial applications: building and optimizing existing deep learning models by leveraging state-of-the-art algorithms to improve performance and accuracy.
- Studying and utilizing machine learning frameworks tailored for computer vision tasks, ensuring efficient implementation and scalability.
- Researching and implementing advanced object tracking algorithms to address specific industrial challenges.
- Implementing object tracking and computer vision techniques to improve system performance.
- Collaborating with team members to integrate optimized models into operational systems and evaluate their performance in production environments.

Computer Science Teacher
ISIS Paschini Linussio, ISIS Raimondo D'Aronco, (Udine), ITA 2023/10 → 2024/09

- Taught programming and computer science to high school students.
- Designed lessons integrating theory and practical exercises.

Computer Vision Engineer (Intern)
Danieli Research Center, Buttrio (Udine), ITA 2022/10 → 2022/12

- Applied computer vision techniques, including calibration methods such as homography estimation for camera alignment, object detection, and object localization, with a focus on real-world industrial challenges.
- Researched and implemented state-of-the-art algorithms to optimize detection and localization of objects in a steel slab storage yard.
- Designed, implemented, and tested convolutional neural networks (CNNs) for object localization and detection.
- Analyzed model performances with hyperparameter tuning methods to identify and deploy the most effective solutions in a production environment.

Education

Master's Degree in Data Science and Scientific Computing
University of Trieste, (Trieste), ITA 2023/10 → Present

Course is jointly offered by the *University of Trieste*, *University of Udine*, *ICTP*, and *SISSA*.

- Key courses: *Advanced Programming and Algorithmic Design, Data Management for Big Data, High Performance Computing, Numerical Analysis, Probabilistic Machine Learning, Reinforcement Learning, Statistical Methods for Data Science, Deep Learning, Information Retrieval and Data Visualization.*

Bachelor's Degree in Internet of Things, Big Data & Web
University of Udine, (Udine), ITA 2019/10 → 2022/12

Final Grade: 100/110 — Thesis: *Applications of Computer Vision for Object Detection and Localization in a Steel Yard Plant.*

- Key courses: *Algorithms and Data Structures, Machine Learning, Data Science, Applied Statistics, Social Computing, Mathematical Analysis, Object-Oriented Programming, Cloud Computing, Software Engineering, Databases.*

Projects

GAPC - Asteroid Observation Catalog: django-based application to catalog asteroid observations, following astronomical standards. It enables researchers to store and analyze asteroid metadata efficiently.

YOHO24: Deep learning model for sound event detection, inspired by YOHO. Developed as part of a Deep Learning course at the University of Trieste.

Weighted Matrix Factorization Model: developed a recommender system using matrix factorization techniques. Implemented and compared optimization methods such as Stochastic Gradient Descent and Weighted Alternating Least Squares (WALS) to enhance recommendation accuracy and system efficiency.

Traffic Lights Optimization: Applied Reinforcement Learning techniques, including Policy Iteration and Value Iter- ation, to optimize traffic light policies, aiming to reduce waiting times and queue lengths.

Languages

Italian: Native speaker.
English: B2 (Listening, Reading, Writing).

Driving License

Category B.